

# Executive Business Report

## Intelligent Sales Forecasting and Demand Intelligence System

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### Executive Summary

This project was developed to help retail managers improve inventory planning and sales forecasting using historical sales data. The system analyzes sales trends, compares multiple forecasting models, detects unusual sales patterns, and groups products based on demand behavior. Three forecasting approaches—SARIMA, Prophet, and XGBoost—were evaluated, with XGBoost delivering the best prediction accuracy. An interactive Streamlit dashboard was also developed to visualize sales trends, forecasts, anomalies, and demand segments, enabling business managers to make informed stocking and planning decisions without requiring technical knowledge.

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### Key Findings from Exploratory Data Analysis and Forecasting

Historical sales data was analyzed to understand business performance across multiple years. Time-series decomposition and trend analysis showed that sales vary over time and contain both trend and seasonal patterns. Stationarity testing and preprocessing were performed before developing forecasting models.

Three forecasting models were evaluated:

| Model          | MAE              | RMSE             | MAPE          |
|----------------|------------------|------------------|---------------|
| SARIMA         | 20,580.70        | 22,190.91        | 21.94%        |
| Prophet        | 40,970.33        | 53,868.95        | 40.04%        |
| <b>XGBoost</b> | <b>14,870.76</b> | <b>18,543.80</b> | <b>14.59%</b> |

XGBoost achieved the lowest forecasting error across all evaluation metrics and was therefore selected as the production model.

### Three-Month Sales Forecast

Based on the XGBoost forecasting model, expected sales for the next three months are:

| Forecast Period | Expected Sales   |
|-----------------|------------------|
| Month 1         | <b>87,144.48</b> |
| Month 2         | <b>87,673.21</b> |
| Month 3         | <b>87,681.38</b> |

**Forecast Confidence (Business Interpretation)**

The forecast indicates relatively stable sales over the next three months, remaining close to **87,000–88,000**. Actual sales may differ because of seasonal promotions, market demand changes, supply chain disruptions, or external economic conditions. However, the model suggests that no major increase or decline in sales is expected during the forecast period.

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**Top Anomalies Detected**

Sales anomalies were detected using **Isolation Forest** and validated using the **Z-Score** method.

The detected anomalies most likely represent:

- 1. **Unusually high sales weeks**, which may correspond to promotional campaigns, festive sales, or bulk customer orders.
  - 2. **Unusually low sales weeks**, which may indicate reduced customer demand, inventory shortages, or operational disruptions.
  - 3. **Weeks identified by both anomaly detection techniques**, representing high-confidence unusual sales events that should be reviewed by business managers for operational insights.
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**Product Demand Segmentation**

Products were segmented using **K-Means Clustering** based on sales volume, sales growth, average order value, and sales volatility.

The resulting demand segments support inventory planning as follows:

| Demand Segment | Recommended Stocking Strategy                                                 |
|----------------|-------------------------------------------------------------------------------|
| High Volume    | Maintain high inventory with frequent replenishment to avoid stock shortages. |
| Growing Demand | Gradually increase inventory to support increasing customer demand.           |
| Stable Demand  | Maintain regular inventory levels with periodic monitoring.                   |
| Low Volume     | Reduce inventory levels and avoid overstocking to minimize holding costs.     |

This segmentation enables more efficient inventory allocation and improves warehouse utilization.

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**Business Recommendations**

### **1. Use XGBoost for Sales Forecasting**

XGBoost produced the lowest forecasting error (**MAE = 14,870.76, MAPE = 14.59%**) among all evaluated models. Using this model for demand planning can improve procurement decisions and inventory management.

### **2. Monitor Sales Anomalies Regularly**

The anomaly detection system automatically identifies unusual sales behavior. Investigating these events can help management understand the impact of promotions, operational issues, or unexpected market changes.

### **3. Optimize Inventory Using Demand Segmentation**

Inventory policies should be based on demand segments rather than treating all products equally. High-demand products require frequent replenishment, while low-demand products should be stocked conservatively to reduce inventory costs.

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### **Risk / Limitation**

The forecasting model is trained using historical sales data. Unexpected events such as economic changes, competitor activities, supply chain disruptions, or large promotional campaigns may affect future sales and reduce forecasting accuracy. The forecasting model should therefore be retrained periodically using updated business data.

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### **Conclusion**

The Intelligent Sales Forecasting and Demand Intelligence System successfully combines forecasting, anomaly detection, demand segmentation, and interactive visualization into a single business solution. After comparing three forecasting techniques, XGBoost was selected as the most accurate model for future sales prediction. The developed dashboard enables business users to monitor historical performance, forecast future demand, detect unusual sales behavior, and optimize inventory planning through data-driven decision-making.